

DQ Decoupling Guide

Power Application Controller™



www.qorvo.com

Copyright © 2023 Qorvo, Inc.

No portion of this document may be reproduced or reused in any form without Qorvo's prior written consent



Table of Contents

1 Description	3
2 Firmware Flow.....	4
3 Using DQ Decoupling.....	5
3.1 Enabling DQ Decoupling	5
3.2 Firmware Options	5
3.3 Firmware Parameters	5
3.4 Tuning Parameters (GUI)	5
Contact Information.....	6
Important Notice	6



1 DESCRIPTION

For a PMSM motor, the voltage equations based on the DQ frame of reference are:

$$V_d = R_s i_d + L_d p i_d - \omega L_q i_q$$

$$V_q = R_s i_q + L_q p i_q + \omega L_d i_d + \omega \Phi$$

The d-axis and q-axis cross-coupling terms $-\omega L_q i_q$ and $+\omega L_d i_d$ respectively are often ignored in the current control loops. Doing so may be inconsequential in many circumstances. But there are some applications where the current control loop requires a very dynamic and fast response. In such cases, compensating for these terms reduces the dependence on the bandwidth of the PI controller to respond to these fast changes. This compensation in the current control system can be accomplished by using a DQ-decoupling technique. The goal here is to make the controllers for each axis independent from each other, so that they don't have to respond to changes in each other's axis and just focus on responding to changes in their own axis.

Additionally, the BEMF term ($\omega \Phi$) can also be compensated for and by doing so the Q-axis current controller would entirely focus on the dynamic changes.

2 FIRMWARE FLOW

The FOC firmware has two options to choose for the DQ decoupling implementation. The two options are:

- 1) Feedforward decoupling
- 2) Feedback decoupling

The basic implementation for these decoupling methods is illustrated in Figure 1 and Figure 2.

If Feedback decoupling is selected, a low-pass filter is applied to the feedback current in order to avoid issues resulting from the injection of high frequency elements into the current control loop.

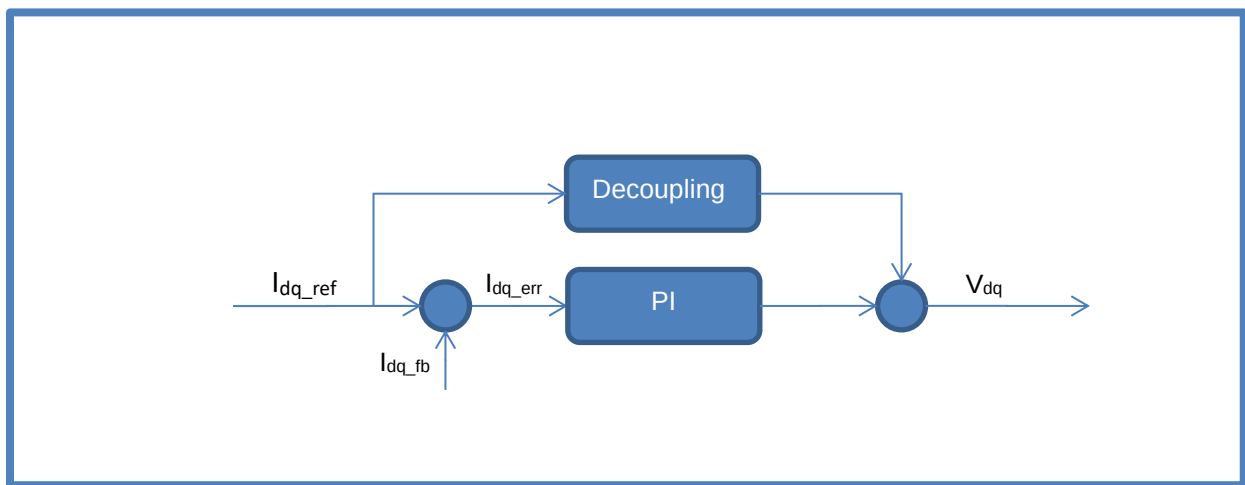


Figure 1: Feedforward Decoupling

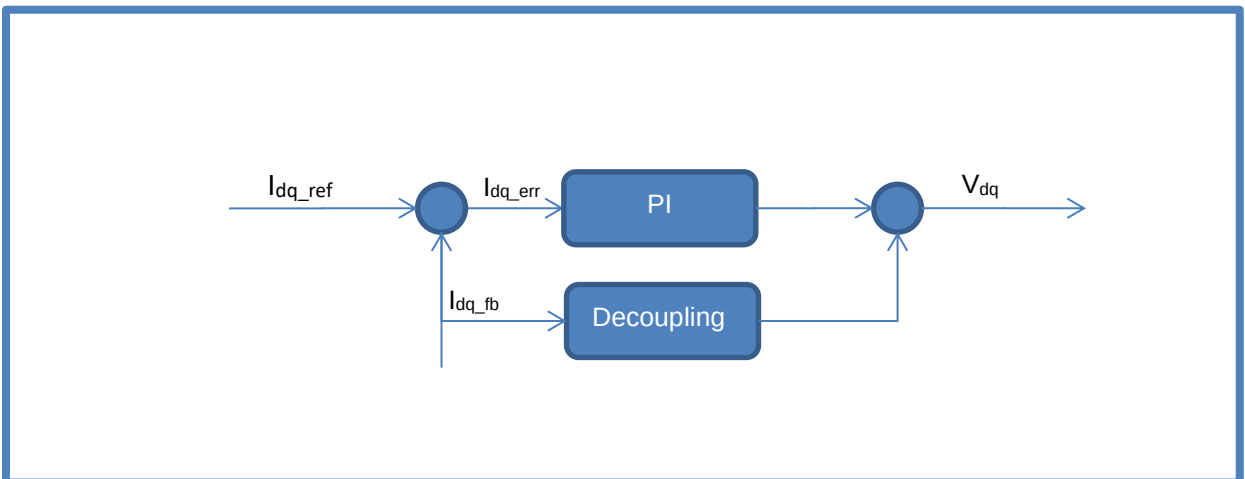


Figure 2: Feedback Decoupling



3 USING DQ DECOUPLING

3.1 Enabling DQ Decoupling

To enable the DQ Decoupling feature, find the compile option in config_features.h and verify that it is defined:

```
#define ENABLE_DQ_DECOUPLING
```

3.2 Firmware Options

To configure DQ Decoupling, please verify that the desired options are defined in config_features.h and that the selections and values are appropriate for the application:

DQ Decoupling Method Selection

```
SEL_DQ_DECOUPLING_METHOD
```

Use to select DQ Decoupling method. The available options are:

```
FEEDFORWARD_DECOUPLING
```

```
FEEDBACK_DECOUPLING
```

3.3 Firmware Parameters

DQ Decoupling uses the firmware specified configuration parameters below. These parameters are defined in config_features.h:

```
CURRENT_FILTER_CUTOFF_FREQ_Q16
```

Maximum cutoff frequency (Hz) for low-pass filter applied to motor current for the decoupling calculation. The filter cutoff frequency is calculated based on the current PI controller bandwidth unless it exceeds this maximum setting.

3.4 Tuning Parameters (GUI)

There are no GUI configurable parameters for DQ Decoupling.



CONTACT INFORMATION

For the latest specifications, additional product information, worldwide sales and distribution locations:

Web: www.qorvo.com

Tel: 1-844-890-8163

Email: customer.support@qorvo.com

IMPORTANT NOTICE

The information contained herein is believed to be reliable; however, Qorvo makes no warranties regarding the information contained herein and assumes no responsibility or liability whatsoever for the use of the information contained herein. All information contained herein is subject to change without notice. Customers should obtain and verify the latest relevant information before placing orders for Qorvo products. The information contained herein or any use of such information does not grant, explicitly or implicitly, to any party any patent rights, licenses, or any other intellectual property rights, whether with regard to such information itself or anything described by such information. **THIS INFORMATION DOES NOT CONSTITUTE A WARRANTY WITH RESPECT TO THE PRODUCTS DESCRIBED HEREIN, AND QORVO HEREBY DISCLAIMS ANY AND ALL WARRANTIES WITH RESPECT TO SUCH PRODUCTS WHETHER EXPRESS OR IMPLIED BY LAW, COURSE OF DEALING, COURSE OF PERFORMANCE, USAGE OF TRADE OR OTHERWISE, INCLUDING THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE.**

Without limiting the generality of the foregoing, Qorvo products are not warranted or authorized for use as critical components in medical, life-saving, or life-sustaining applications, or other applications where a failure would reasonably be expected to cause severe personal injury or death.

Copyright 2019 © Qorvo, Inc. | Qorvo is a registered trademark of Qorvo, Inc.